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# Missing Mandates for the standard library

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## Introduction

As the "Mandating the standard library" papers have progressed/been applied, a few bits have been missed; mostly because parts of the library have moved.

This paper fixes those bits.

Thanks to Richard Smith for enumerating these parts.

## Wording

These changes are based on [N4842](#).

In [tab:cpp17 allocator], change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ `(*p).m` is well-defined.

Preconditions:~~Requires:~~ `(*q).m` is well-defined.

Preconditions:~~Requires:~~ `p` ~~is~~~~shall be~~ a value returned by an earlier call to `allocate` that has not been invalidated by an intervening call to `deallocate`. `n` ~~matches~~~~shall match~~ the value passed to `allocate` to obtain this memory.

In [res.on.required]/1 add:

Violation of any preconditions specified in a function's Preconditions: ~~or~~ *Requires:* element results in undefined behavior

In [coroutine.handle.con]/2, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ `p` is a reference to a promise object of a coroutine.

In [coroutine.handle.export.import]/2, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ addr was obtained via a prior call to address

In [coroutine.handle.observers]/2 and /4, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ \*this refers to a suspended coroutine

Preconditions:~~Requires:~~ \*this refers to a suspended coroutine

In [coroutine.handle.promise]/1, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ \*this refers to a coroutine.

In [util.smartptr.atomic.shared]/3, /6, and /12, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ order is neither~~The order argument shall not be~~ memory\_order::consume, memory\_order::acquire, nor memory\_order::acq\_rel.

Preconditions:~~Requires:~~ order is neither~~shall not be~~ memory\_order::release nor memory\_order::acq\_rel.

Preconditions:~~Requires:~~ failure is neither~~shall not be~~ memory\_order::release nor memory\_order::acq\_rel.

In [util.smartptr.atomic.weak]/6 and /12, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ order is neither~~shall not be~~ memory\_order::release nor memory\_order::acq\_rel

Preconditions:~~Requires:~~ failure is neither~~shall not be~~ memory\_order::release nor memory\_order::acq\_rel